



Exercise Session 9 Highlights

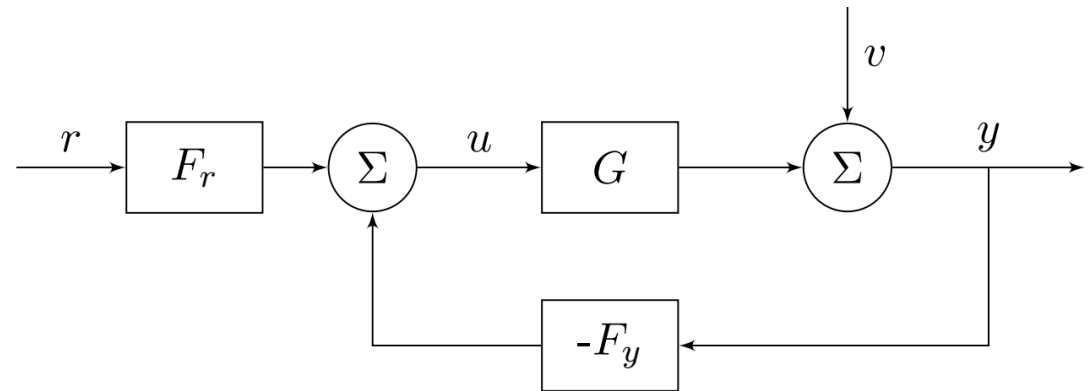
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Course: Reglerteknik (EL1000)

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Review of Sensitivity Function

Feedback reduced the effect of the external disturbance signal v on the system's output y



$$Y = \frac{GF_r}{1 + GF_y} R + \frac{1}{1 + GF_y} V$$

Sensitivity Function: (Transfer function from external disturbance to the system output)

$$S(s) = \frac{1}{1 + G(s)F_y(s)}$$

We get little impact from disturbances with frequencies where $|S(i\omega)|$ is small

Review of Robustness Criterion

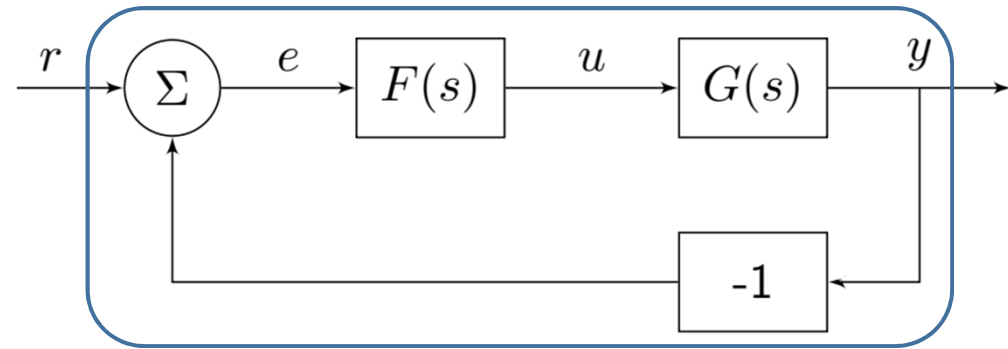
Nominal system: $G(s)$

In reality, the system behaves as:

$$G^0(s) = G(s) [1 + \Delta G(s)]$$

The relative error between the real system and the nominal model:

$$\Delta G(s) = \frac{G^0(s) - G(s)}{G(s)}$$



$$G_c = \frac{GF}{1 + GF}$$

Under what condition the real system is stable if the nominal system is already stable?

$$|T(i\omega)| < \frac{1}{g(\omega)} \quad \forall \omega$$

$$\text{where } T = G_c \text{ and } |\Delta G(i\omega)| \leq g(\omega) \quad \forall \omega$$

More precisely when:

$$\frac{|F(i\omega)G(i\omega)|}{|1 + F(i\omega)G(i\omega)|} < \frac{1}{|\Delta G(i\omega)|} \quad \forall \omega$$

State-Space Equations

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = b_0u + b_1\dot{u} + b_2\ddot{u}$$

Idea: an n -th order linear differential equation can be represented by n first order linear differential equations

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

$x(t)$ = State vector ($n \times 1$ dimension);

$y(t)$ = Output vector ($p \times 1$ dimension);

$u(t)$ = Input vector ($r \times 1$ dimension);

A = State matrix ($n \times n$ dimension);

B = Input matrix ($n \times r$ dimension);

C = Output matrix ($p \times n$ dimension);

D = Direct transmission matrix ($p \times r$ dimension).

In this course, since we often deal with single input single output systems we have $p = 1$ and $r = 1$

State-Space Equations to Transfer Function

By application of the Laplace transformation to the state space equation one can show:

$$Y(s) = G(s)U(s), \quad G(s) = C(sI - A)^{-1}B + D$$

$$G(s) = \frac{\overbrace{C(sI - A)^*}^{\text{adjugate matrix}} B}{\underbrace{\det(sI - A)}_{\rightarrow \det(sI - A) = 0 \text{ gives the eigenvalues of matrix } A}} + D$$

Eigenvalues of matrix A represent the poles of the system!

Transfer Function to State Space form

There are basically an **infinite** number of state space descriptions for one system, depending on how the state vector is chosen!

Some famous forms:

$$G(s) = \frac{b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

State Space in Controllable Canonical form:

$$\dot{x} = \begin{bmatrix} -a_1 & -a_2 & \dots & -a_{n-1} & -a_n \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$
$$y = [b_1 \quad b_2 \quad \dots \quad b_n] x$$

Transfer Function to State Space form (cont.)

$$G(s) = \frac{b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

State Space in Observable Canonical form:

$$\dot{x} = \begin{bmatrix} -a_1 & 1 & 0 & \dots & 0 \\ -a_2 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_{n-1} & 0 & 0 & \dots & 1 \\ -a_n & 0 & 0 & \dots & 0 \end{bmatrix} x + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} x$$

Linearization

General nonlinear system dynamics in state space form:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= h(x, u).\end{aligned}$$

We can linearize such a system around a stationary (equilibrium) point to approximate the system behavior by a linear function that is only valid around the stationary point.

At the stationary (equilibrium) point (x_0, y_0, u_0) we have (system is at rest, i.e., $\dot{x} = 0$):

$$\boxed{f(x_0, u_0) = 0} \quad \rightarrow \quad y_0 = h(x_0, u_0)$$

application of
Taylor expansion

$$\begin{aligned}x &= x_0 + \Delta x \\ u &= u_0 + \Delta u\end{aligned}$$

variations

$$\begin{aligned}\Delta \dot{x} &= A\Delta x + B\Delta u \\ \Delta y &= C\Delta x + D\Delta u\end{aligned}$$

linear dynamics
approximating the
behaviour of nonlinear
dynamics around
 (x_0, u_0)

Linearization (cont.)

Matrices A , B , C , and D are given by the Jacobians of f and h as follows (from Taylor expansion):

$$A = f_x(x_0, u_0) = \left[\begin{array}{ccc} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{array} \right] \bigg|_{(x_0, u_0)}$$

$$B = f_u(x_0, u_0) = \left[\begin{array}{c} \frac{\partial f_1}{\partial u} \\ \vdots \\ \frac{\partial f_n}{\partial u} \end{array} \right] \bigg|_{(x_0, u_0)}$$

$\leadsto$ if u is scalar

for $u = [u_1, u_2]^T$

$$B = \left[\begin{array}{cc} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \vdots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} \end{array} \right] \bigg|_{(x_0, u_0)}$$

$$C = h_x(x_0, u_0) = \left[\begin{array}{ccc} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_n}{\partial x_n} \end{array} \right] \bigg|_{(x_0, u_0)}$$

$$D = h_u(x_0, u_0) = \frac{\partial h}{\partial u} \bigg|_{(x_0, u_0)}.$$

References

- Lectures of Automatic Control, General Course (EL1000), KTH, 2021
- Rebecka Winqvist's notes for Automatic Control, General Course (EL1000), KTH, 2021-2022

Useful Videos on State Space & Linearization (Optional):



You can also watch **Brian Douglas'** (newer) videos:
(these videos might help you to better grasp the concepts but are not designed for solving exercises)



Intro to state space models



Intro to linearization